Mehreen Bintay Yousuf Simy
20254103266, Department of Computer Science and Engineering (CSE)
Bangladesh University of Business and Technology
Mehreensimin86@gmail.com

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The Head of the Department
Department of Computer Science and Engineering (CSE)
Bangladesh University of Business and Technology

Subject: Submission of Technical Proposal for "SmartCSE Assist"

Dear Sir/Madam,

I am pleased to submit the technical document, titled "SmartCSE Assist: A Proposal for an AI-Powered Student Support Application," as requested by the department. The document is made for presentating to a software/AI company for possible development support.

The proposal describes a mobile application that uses artificial intelligence to support students in areas like: personalised course recommendations, assignment reminders, basic programming help, and answers to frequently asked academic questions. It outlines the system workflow, compares four AI-based features by input data, technique, and expected benefit, and discusses limitations, including privacy and answer reliability. The statements in the document are supported by references to published literature.

The proposed application is valuable because students must currently track courses, deadlines, and routine queries through separate channels. A single application with limited, consent-based data use could make academic planning easier and reduce repetitive queries to staff. The expected benefits would, of course, need to be confirmed through evaluation.

I would be grateful for your review and any feedback on the proposal. Thank you for your time and consideration.

Yours sincerely,

Mehreen Bintay Yousuf Simy
Student, Department of CSE

SmartCSE Assist: A Proposal for an AI-Powered Student Support Application

Mehreen Bintay Yousuf Simy | Department of Computer Science and Engineering

1.Abstract

This document proposes SmartCSE Assist, a mobile application that uses artificial intelligence to provide personalised course recommendations, assignment reminders, basic programming support, and answers to frequently asked academic questions. The system relies on limited student data, including courses, interests, and activity history. The expected benefits are to improved academic planning, faster access to support, and reduced student workload, while privacy is protected through data minimisation.

2.Introduction and Problem statement

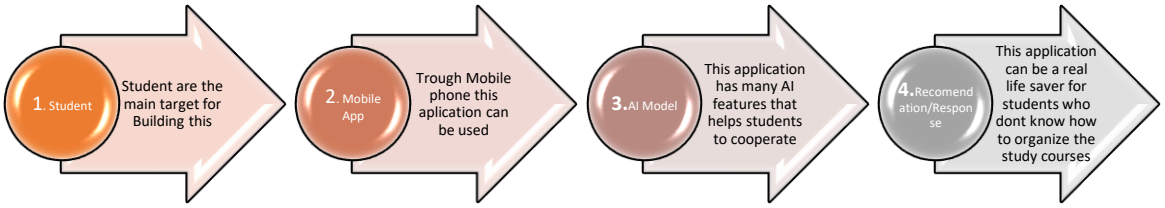
CSE students manage several courses, frequent assignment deadlines, and elective choices at the same time. Reviews of AI in higher education report growing use of adaptive systems, student profiling, and intelligent tutoring to support such needs. In many departments, however, students still rely on notices, informal advice, and manual tracking. This can lead to missed deadlines, poorly matched course selections, and delayed answers to routine questions.

The department therefore proposes SmartCSE Assist, a mobile application that applies AI to these problems. The application is built around three core AI-based features. First, a course recommender suggests courses from a student's completed courses, interests. Second, a conversational assistant answers frequently asked academic questions and gives basic programming help using natural language. Third, a reminder module assignment deadlines according to the student's schedule.

3.Proposed System

- "SmartCSE Assist"The mobile application will use Artificial Intelligence to help students with academic planning, course
- recommendations, assignment reminders, basic programming support, and frequently asked academic questions.
- The proposed app will collect limited student information such as courses, academic interests, and activity history
- to provide personalized recommendations. The department wants to submit a short technical proposal to a
- software/AI company for possible development support.

4.System Workflow



The proposed workflow is shown in Figure-1

5.Features Details

AI Features	input/data	AI technique	Privacy/Machine learning
Course recomendation	Complete course, Academic interest, activities history	Content-based and collaborative filtering; matrix factorisation	Ranked list of suitable courses to support elective planning
Academic Assistent	Get immediate Answers by the student queries	Transformer-based language model with retrieval from verified sources	Prompt, consistent answers; fewer routine queries to staff
Basic Programming Support	Help in coding and debugging program	Transformer-based language model for error explanation and hints	Plain-language explanations and hints for debugging
Smart Assignment Reminder	Reminder for newly added or previous assignments before deadline	Priority scoring with rule-based scheduling	Timely, prioritised notifications to reduce missed deadlines

Table 1 summarizes the major AI features.

6.Expected Benefits and Limitation

The main expected benefits are personalised guidance, access to support outside office hours, and a lower volume of repetitive queries to academic staff. AI applications in education have been associated with such personalisation and administrative support

Several limitations must be addressed. Recommender systems perform poorly for new students with little history, a problem known as cold start, and for sparse data. Language models can generate incorrect statements, so the assistant should answer only from verified department sources and refer students to staff for official decisions. The programming assistant should offer explanations and hints rather than complete solutions, to protect academic integrity. Finally, any storage of student data creates privacy and security risks that require access control, encryption, and consent management.

7.Conclusion

SmartCSE Assist proposes a single mobile application that combines course recommendation, an assistant, programming support, and smart reminders using limited student data. The design relies on established AI techniques and includes safeguards for privacy and answer reliability. A pilot study with CSE students is recommended before full deployment to measure whether the expected benefits are achieved.

References

[1] O. Zawacki-Richter, V. I. Marín, M. Bond, and F. Gouverneur, "Systematic review of research on artificial intelligence applications in higher education – where are the educators?," Int. J. Educ. Technol. High. Educ., vol. 16, no. 1, Art. no. 39, 2019.

[2] L. Chen, P. Chen, and Z. Lin, "Artificial intelligence in education: A review," IEEE Access, vol. 8, pp. 75264–75278, 2020.

[3] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," Computer, vol. 42, no. 8, pp. 30–37, Aug. 2009.

[4] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, "Recommender systems survey," Knowl.-Based Syst., vol. 46, pp. 109–132, 2013.

[5] A. Vaswani et al., "Attention is all you need," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017.